



QUANTITATIVE METHODS 1 · CHAPTER 6

Statistical Index Numbers

Exercise series

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1. Elementary index and rebasing

A service price rises from EUR 40 in 2023 to EUR 46 in 2025.

- 1 Calculate the 2025 index with 2023=100 and the percentage change.
- 2 An old series is 90, 100, 108 for 2022-2024 with 2023=100. Rebase it to 2022=100.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Value, Laspeyres and Paasche

Product	p_0	q_0	p_1	q_1
A	2	100	2.5	90
B	5	40	6	50

Calculate the value index and Laspeyres and Paasche price indices.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Fisher and factor reversal

Using Exercise 2, calculate Fisher's price index, Laspeyres and Paasche quantity indices, Fisher's quantity index and verify factor reversal.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. Fixed-base, chain and deflation

An index is 100 in year 0 and 110 in year 1. The year-2 link index relative to year 1 is 106. Nominal year-2 revenue is EUR 636,000.

- 1 Construct the chain index for year 2.
- 2 Deflate revenue into year-0 euros.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. Supplier dashboard decision

A buyer reports: value index 128, Fisher price index 120 and Fisher quantity index 106.67. Explain the three numbers, verify consistency, and give two cautions before attributing the price rise to supplier market power.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Complete price-quantity decomposition

A purchasing department observes:

Product	p0	q0	p1	q1
A	4	100	5	90
B	10	50	11	60
C	20	20	18	30

- 1 Calculate the four totals $\Sigma p_0 q_0$, $\Sigma p_1 q_0$, $\Sigma p_0 q_1$ and $\Sigma p_1 q_1$.
- 2 Calculate the value, Laspeyres-price, Paasche-price and Fisher-price indices.
- 3 Interpret the gap between value and price change.

Method prompt. Complete all four cross-value columns before forming any index.

7. Rebasing, inflation and real performance

A nominal revenue series is €500k, €560k and €630k. The corresponding price index (year 1 = 100) is 100, 108 and 120.

- 1 Calculate real revenue in year-1 euros.
- 2 Calculate nominal and real growth from year 1 to year 3.
- 3 Rebase the price index so year 2 = 100.
- 4 Explain the result to a manager who claims that activity grew by 26%.

Method prompt. Deflate by dividing nominal value by index/100. Rebasing changes the reference, not the underlying inflation path.